

Coffee Shop Sales Analysis: Descriptive Insights and Predictive Modeling

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Abstract

This report presents a comprehensive analysis of coffee shop transaction data, combining exploratory data analysis (EDA) with predictive modeling. The primary goal is to uncover sales patterns, identify key temporal trends, and evaluate the feasibility of forecasting daily sales. While descriptive insights reveal clear product preferences and peak business hours, predictive models—including SARIMA and XGBoost—struggle to outperform a simple persistence baseline, indicating high day-to-day volatility. The findings offer actionable business recommendations for inventory planning and product placement, while highlighting the need for additional features to improve forecast accuracy.

1 Introduction

Understanding customer behavior and sales dynamics is essential for optimizing inventory, staffing, and marketing strategies in a coffee shop. This notebook analyzes transaction records over five months (March–July 2024) to:

- Identify the most popular products and revenue drivers.
- Uncover temporal patterns across months, weekdays, and hours.
- Develop and compare time series forecasting models for daily sales.

2 Dataset Overview

The dataset contains 1,133 transactions with the following attributes:

- `date` and `datetime` of purchase.
- `cash_type` (card / cash) and anonymized `card` number.
- `money` (transaction amount).
- `coffee_name` – one of eight coffee products.

Missing values in the `card` column correspond to cash payments (89 records). No duplicate entries exist.

3 Exploratory Data Analysis

3.1 Payment Methods

Approximately 92% of transactions are paid by card, indicating a strong preference for cashless payments.

3.2 Product Popularity

Coffee Name	Revenue Contribution (%)
Americano with Milk	23.6
Latte	21.4
Cappuccino	17.3
Americano	14.9
Cortado	8.7
Hot Chocolate	6.5
Espresso	4.3
Cocoa	3.1

Table 1: Relative revenue share by coffee type.

Latte generates the highest revenue, while Americano with Milk is the most frequently purchased.

3.3 Temporal Patterns

- **Monthly trends:** Sales peak in May and June for most products; Americano with Milk shows a sharp increase in June.
- **Weekday patterns:** Tuesday is the busiest day (185 transactions), while Sunday and Monday have the lowest counts.
- **Hourly patterns:** A pronounced peak occurs at 10:00 AM, especially for Latte. Cappuccino, Cocoa, and Hot Chocolate also see secondary peaks in the evening (6–8 PM).

4 Predictive Modeling

4.1 Data Preparation

Daily total sales were aggregated from the transaction records, resulting in 150 days of data. Features engineered include:

- Temporal indicators: day of week, month, day of month, week of year.
- Lagged sales (1, 2, 3, 7, 14 days).
- Rolling statistics (7-day mean and standard deviation).

Data was split 80/20 into training (2024-03-15 to 2024-07-03) and testing (2024-07-04 to 2024-07-31) sets.

4.2 Models Evaluated

1. **Persistence**: naive forecast using the previous day’s sales.
2. **SARIMA**: auto-selected seasonal ARIMA model (order (0,1,3)(0,0,1)[7]).
3. **XGBoost**: gradient boosting with 100 trees, learning rate 0.1.
4. **Ensemble**: simple average and weighted average (weights inverse to MAE) of SARIMA and XGBoost.

4.3 Results

Performance on the test set is summarized in Table 2.

Model	MAE	RMSE	R ²	MAPE (%)	Acc. ($\pm 20\%$)
Persistence	4.25	5.47	0.121	89.9	25.0
SARIMA	4.53	6.11	-0.098	93.5	10.7
XGBoost	4.75	6.07	-0.085	125.9	17.9
Ensemble (Avg)	4.63	6.03	-0.068	109.6	10.7
Ensemble (Wtd)	4.63	6.03	-0.069	109.2	10.7

Table 2: Forecast performance metrics on the test set.

4.4 Discussion

The persistence model achieves the lowest MAE (4.25), confirming that daily sales exhibit strong short-term autocorrelation. SARIMA and XGBoost produce higher MAE and negative R², indicating that they fail to capture day-to-day variability better than the naive baseline. The ensembles do not improve performance, as both constituent models are already weak. The extremely high MAPE values (especially for XGBoost) reflect the presence of several outlier days where predictions deviate substantially.

Figure 1 illustrates the actual vs. predicted sales for the test period. XGBoost tracks the general trend but overshoots some peaks; SARIMA is smoother and misses sharp fluctuations.

Figure 1: Actual vs. predicted daily sales. XGBoost (blue) and SARIMA (red) predictions are shown alongside the actual values (black).

5 Conclusion

The exploratory analysis successfully identified key sales drivers: Latte is the top revenue generator, Tuesday is the busiest weekday, and 10AM is the peak hour for most products. These insights can directly inform inventory management, product placement, and staffing decisions.

However, predictive modeling proved challenging. The persistence baseline outperformed all statistical and machine learning models, underscoring the inherent volatility of daily sales. The high MAPE and negative R^2 suggest that daily forecasts are unreliable with the current feature set. To improve prediction accuracy, future work should incorporate external factors such as weather data, holidays, and local events, or aggregate sales to a weekly level to reduce noise. Nevertheless, the descriptive findings already provide actionable business intelligence for the coffee shop.